DIGI-GUARD

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Chapter 1

Introduction

Identify the product or application whose requirements are specified in this document. Describe the different types of reader that the document is intended for, such as developers, project managers, marketing staff, users, testers, and documentation writers. [1].

1.1 Problem Statement

In this digital world, children have access to all kinds of smartphones, tablets, and the internet. While this access provides a lot of educational and social opportunities, it also exposes them to inappropriate content and cyberbullying. Parents have many concerns about how to protect their children from these kinds of potential dangers, and they often lack the necessary tools to control and manage their children's online activities. However, there are many existing solutions, but they are usually either too complicated, lack comprehensive features, or don't have real-time monitoring. The primary problem is that parents need software that will provide them detailed real-time insights into their children's activities and allow them to restrict their access to certain content. Moreover, parents also want to manage the time their children spend on devices to ensure a balance between digital and other life activities. In addition, location tracking is essential for parents to ensure the physical safety of their children.

1.2 Scope

The project is about building a Parental Control App that facilitates the parents' management and tracking of their children's digital activities. This solution provides real-time monitoring, location tracking, app restrictions, and screen time management to enable a safe digital experience for kids. The project focuses on creating a very friendly platform for parents to control and safeguard the digital presence of their child. This will block/limit indecent content, monitor real-time device usage, and trace a child's location for the purpose of enhanced security. Additionally, you can remotely control the device's microphones and cameras, keylog, and know the child's WhatsApp location, providing a whole-home parental control solution. The project's scope is defined by focusing on functions like app restrictions, screen time management, location tracking, and real-time monitoring, which ensures the safe use of digital platforms for children while giving parents full control over their children's online access.

1.3 Modules

Digiguard is structured into several modules, each responsible for a different aspect of functionality. Below are the key modules of the Android application:

1.3.1 App Restrictions

This module empowers parents by customizing the child's access to apps.

- **App Blocking**: Allows parents to block any application they do not want their child to access.
- **Time-based Restriction**: Specifies the use of a particular app for only a certain period, thereby limiting its total usage.

1.3.2 Screen Time Management

This module allows parents to control and limit total device usage by children.

- **Daily/Weekly Screen Time Limits**: Parents can set screen time limits, and when the limit is exceeded, the app will automatically lock the device.
- **Screen Time Monitoring**: Provides an overview of how much screen time has been used, whether on a daily or weekly basis.

1.3.3 Location Tracking Feature

This module allows parents to track their child's current location and set premises for children.

- **GPS-based Location Tracking**: Displays the child's location on a map within the app.

- **Geofencing**: Enables parents to set restricted areas for the child and receive alerts when the child leaves these areas.

1.3.4 Remote Controls

As a precautionary measure, this module allows parents to take full control of the device when necessary.

- **Remote Device Locking**: Parents can lock the device remotely, making it impossible to use.

- **Remote App Control**: Certain apps can be enabled or disabled remotely based on the parent's preferences.

1.3.5 Real-time Device Monitoring

This module enables parents to track the child's device usage and screen activity in real-time.

- **Live Screen Monitoring**: Parents can view a live feed of the device's screen to see which apps are in use or what their child is up to.

- **Activity Logs**: A log of the child's app usage is available for review.

- **Keylogging**: Logs all keystrokes made on the device, providing parents with detailed insights into typed messages and activities.

1.3.6 Notifications and Alerts

- **Geofence Alerts**: Parents are notified when their children leave or return to the pre-defined safe zone.

- **Screen Time Alerts**: Alerts are generated when the child approaches or exceeds the maximum daily screen time limit.

1.3.7 Emergency SOS Feature

- **Emergency SOS Button**: A practical button that sends an emergency signal via the parent's app containing the child's GPS coordinates.
- **Automatic Location Sharing**: The app starts sending the child's location every minute after the SOS alarm is triggered until the issue is resolved.

1.4 User Classes and Characteristics

User Class	Description
Parent	A parent is the primary user of the app who wants to monitor and control their child's device usage. They will use features such as real-time monitoring, app restrictions, location tracking, and screen time management. Parents may vary in their tech-savviness, with some being very familiar with technology while others may require more assistance in navigating the app.
Child	The child is the secondary user whose device is being monitored. They will have limited access to the app, primarily for receiving alerts (e.g., screen time warnings or emergency notifications). The child's age and understanding of technology will vary, affecting how they perceive the restrictions set by the parent.

Table 1.1: User Classes and Characteristics for the Parental Control App

Chapter 2

Project Requirements

2.1 Use-case Diagram

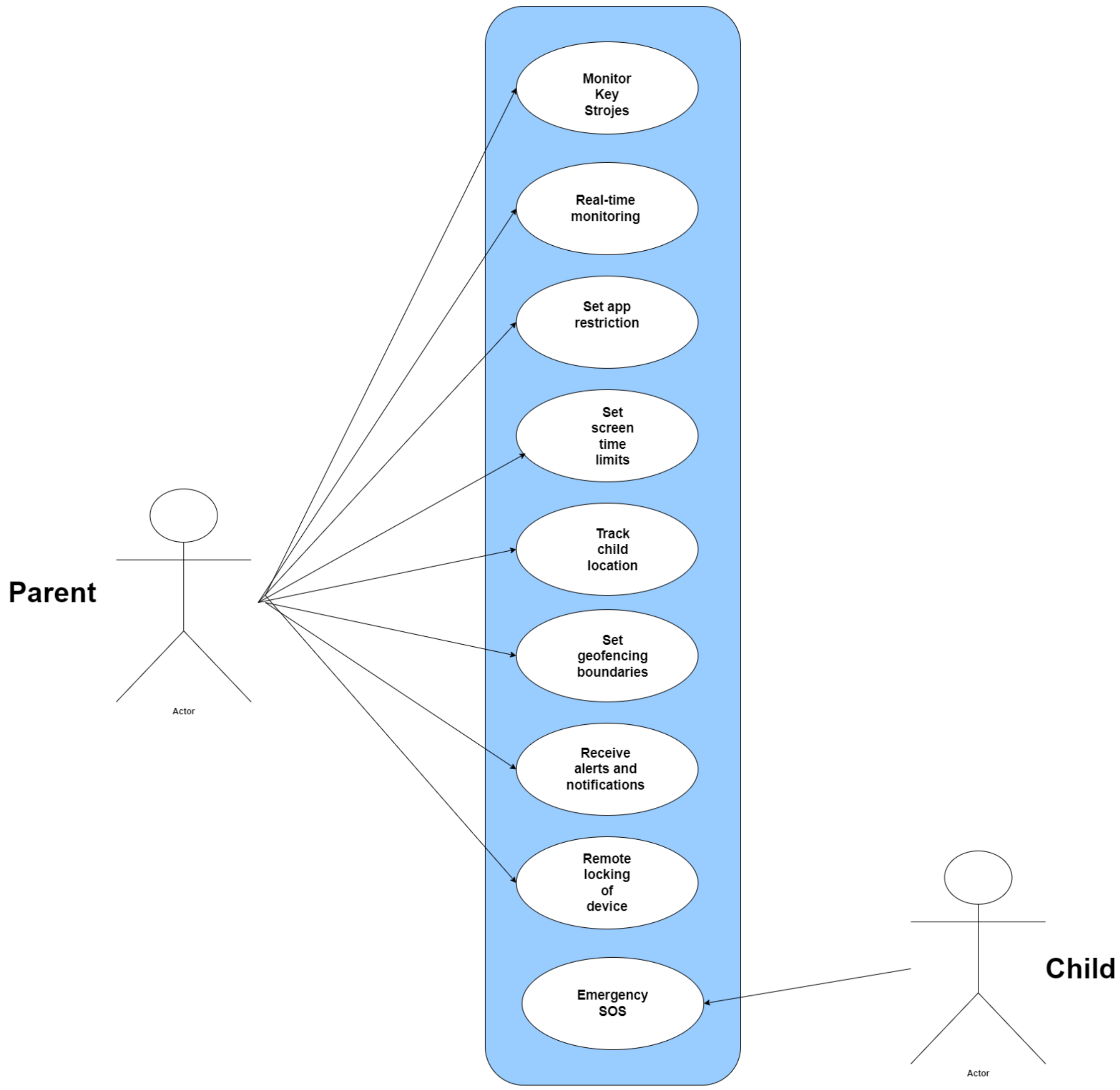

Figure 2.1: Use Case

2.2 Fully Dressed Use Cases

2.2.1 Use Case 1: Monitor Child's Screen (Real-Time Monitoring)

- **Use Case Name:** Monitor Child's Screen
- **Primary Actor:** Parent
- **Secondary Actor:** Child
- **Goal:** To monitor the real-time activity on the child's device.
- **Pre-conditions:**
  - Parent must be logged into the app.
  - Child's device must be connected to the internet and linked to the parent's account.
- **Post-conditions:**
  - Parent can view the child's current screen activity.
  - Parent can choose to log information or restrict apps.
- **Main Success Scenario:**
  1. Parent opens the app and selects "Live Monitoring."
  2. App connects with child's device..
  3. Parent sees what the child is viewing on the screen.
  4. Parent takes further action Parent takes further action (for example, limits the application.
- **Extensions (Alternative Flows):**
  - 3a. Child's device is not online at this time. The app shows: "Device not connected."
  - 3b. Failed to connect. App will try again in a few seconds later.

2.2.2 Use Case 2: Set App Restrictions

- **Use Case Name:** Set App Restrictions
- **Primary Actor:** Parent

- **Secondary Actor:** Child
- **Goal:** The parent will block or restrict the use of particular apps on the child's device.
- **Pre-conditions:**
  - Parent will be authenticated in the application.
  - Child's device is connected and has an active internet connection
- **Post-conditions:**
  - Blocked or time-limited apps.
- **Main Success Scenario:**
  1. The Parent shall select "App restrictions" from the menu in.
  2. App lists installed applications in the child's device
  3. Parent selects the application to be restricted and prescribes a type of restriction or blocking of the app.
  4. The restriction settings are transmitted to the child's device.
  5. The application is blocked or the time is limited.
- **Extensions (Alternative Flows):**
  - 4a. If the restriction fails, the app retries and notifies the parent of failure.
  - 4b. If the app is already restricted, the parent sees: "App already restricted."

2.2.3 Use Case 3: Set Screen Time Limits

- **Use Case Name:** Set Screen Time Limits
- **Primary Actor:** Parent
- **Secondary Actor:** Child
- **Goal:** To limit the times that the child will use the screen for the purposes of playing games, watching movies, or even socializing using the application.
- **Pre-conditions:**
  - Parent logs in to the application. .
  - Child device is online and connected

- **Post-conditions:**
  - Parent limits the child's screen times.
- **Main Success Scenario:**
  1. Parent logs in to the application and selects "Screen Time Limits."
  2. Parent determines the amount of hours eligible to be logged on a daily/ weekly basis.
  3. App enforces those limits on the child device.
  4. Lock the phone or further impose limits on it when screen times are exceeded.
- **Extensions (Alternative Flows):**
  - 3a. If the child exceeds the screen time, the app notifies the parent.

2.2.4 Use Case 4: Track Child's Location

- **Use Case Name:** Track Child's Location
- **Primary Actor:** Parent
- **Secondary Actor:** Child
- **Goal:** To view the child's real-time location on a map.
- **Pre-conditions:**
  - Parent must be logged into the app.
  - Child's device must have location services enabled and be connected to the internet.
- **Post-conditions:**
  - Parent views the child's real-time location and history.
- **Main Success Scenario:**
  1. The parent chooses "Location Tracking" from the app.
  2. The app asks for the child's location data.
  3. Location tracking for the child is shown on the map, for now.
  4. The parent configures geofencing alerts
- **Extensions (Alternative Flows):**

  - 3a. If location services have not been enabled on the child's device then the app shows: "Location services not enabled on the child's device."
  - 3b. If GPS signal is weak, the app uses Wi-Fi or cell tower triangulation.

2.2.5 Use Case 5: Set Geofencing Boundaries

- **Use Case Name:** Set Geofencing Boundaries
- **Primary Actor:** Parent
- **Secondary Actor:** Child
- **Goal:** To set geographic boundaries and receive alerts when the child leaves or enters the geofenced area.
- **Pre-conditions:**
  - Parent must be logged into the app.
  - Child's device must have location services enabled.
- **Post-conditions:**
  - The app sends alerts to the parent when the child crosses the geofenced area.
- **Main Success Scenario:**
  1. Parent accesses the "Geofencing" option through the application.
  2. Parent draws a boundary along the same line on the map.
  3. Boundary is installed by geofence app and child location is tracked.
  4. If the child exits the area, the parent receives an alert.
- **Extensions (Alternative Flows):**
  - 4a. If the geofencing alert fails, it automatically retries in the next possible time.

2.2.6 Use Case 6: Receive Alerts and Notifications

- **Use Case Name:** Receive Alerts and Notifications
- **Primary Actor:** Parent
- **Secondary Actor:** Child

- **Goal:** To receive alerts and notifications when certain predefined conditions are met.
- **Pre-conditions:**
  - Parent must have enabled alerts in the app.
  - The child's device must be linked to the parent's account.
- **Post-conditions:**
  - Parent receives notifications (e.g., screen time exceeded, geofence exit).
- **Main Success Scenario:**
  1. Parent sets up alerts in the app (e.g., screen time limit, geofencing).
  2. The app monitors for those conditions.
  3. When a condition is met, the app sends a notification to the parent.
- **Extensions (Alternative Flows):**
  - 3a. If notifications are delayed, the app queues them for later delivery.

2.2.7 Use Case 7: Remote Lock Device

- **Use Case Name:** Remote Lock Device
- **Primary Actor:** Parent
- **Secondary Actor:** Child
- **Goal:** To remotely lock the child's device.
- **Pre-conditions:**
  - Parent must be logged into the app.
  - Child's device must be online and linked to the parent's account.
- **Post-conditions:**
  - The child's device is locked, preventing further use.
- **Main Success Scenario:**
  1. A parent activates "Remote Lock" from the application
  2. It transmits the locking request of the child's device

3. Child's device locks up

- **Extensions (Alternative Flows):**
  - 3a. In case the child device goes off-line, it places this request in a queue to lock Until device is available again.

2.2.8 Use Case 8: Emergency SOS

- **Use Case Name:** Emergency SOS
- **Primary Actor:** Child
- **Secondary Actor:** Parent
- **Goal:** To send an emergency alert with the child's current location.
- **Pre-conditions:**
  - Child must be able to trigger the SOS feature.
  - Parent must be connected to the child's device.
- **Post-conditions:**
  - Parent receives the child location and an emergency alert.
- **Main Success Scenario:**
  1. Child presses the SOS button in the app.
  2. App sends the child's location to the parent.
  3. Parent receives the SOS notification and location.
  4. The location is updated every 2-3 minutes until the parent responds.
- **Extensions (Alternative Flows):**
  - 2a. If the parent does not respond, the app continues sending updates.

2.2.9 Use Case 9: Monitor Keystrokes and Activity Logs

- **Primary Actor:** Parent
- **Goal:** To log the child's keystrokes and review app usage history.
- **Pre-conditions:**

- Parent must be logged into the app.
- The child's device must be connected and keylogging enabled.

- **Post-conditions:**
  - The application sends keystroke logs and app usage reports back to the parent.

- **Main Success Scenario:**
  1. Parent selects "Key Logs" in the application.
  2. The application retrieves keystroke logs and app usage history.
  3. The parent reviews the history of keys typed and app usage details.

- **Extensions (Alternative Flows):**
  - 3a. If keylogging is disabled, the app prompts the parent to enable it.
  - 3b. If the child's device is offline, the app displays a "Device not connected" message and retries after a short interval.

2.3 Functional Requirements

2.3.1 Module 1: Real-Time Device Monitoring

The requirements for Module 1 are as follows:

1. The application will need to allow the parent to see the live feed of the screen so that it can monitor which apps are accessed or activity that is being conducted

2. The application must provide real-time indications of activity on the screen about the child, showing the running applications at a given time.

3. The app must log the child's app usage for review, storing records of which apps were used and for how long.

4. Allow parents to be able to access the activity log and review the history of app usage.

2.3.2 Module 2: Location Tracking Feature

Requirements for Module 2:

1. This application should be able to notify the parent about the location of the child in a real-time map.
2. The application has to make use of GPS to provide the exact location tracking.
3. The application should provide the parents with the ability to define their geofencing areas, including safe zones where the child is allowed to move around.
4. The application should notify the parent if the child leaves the geofenced area or enters the geofenced area.

2.3.3 Module 3: App Restrictions

Following is the requirement specification for Module 3:

1. App must block applications on the child's device for parents and not accessible by child.
2. The app must implement time-based restrictions, allowing specific apps to be used only within a designated time period
3. The app has to notify in real-time if the child attempts to use a restricted app.
4. The app should record each blocked attempt of an application access made by a child, which the parent needs to view.

2.3.4 Module 4: Screen Time Management

The requirements for Module 4 are as follows:

1. The app must allow setting in-app parental controls for daily or weekly limits on screen time.
2. The app must automatically lock the child's device when the screen time limit has been exceeded.
3. The app must provide an overview of the child's screen usage for a day or week, showing total screen time and app usage.
4. The app should alert the parent that the child is approaching or has exceeded the screen time limit.

2.3.5 Module 5: Remote Controls

Following are the requirements for Module 5:

1. The application should offer parents the options for managing the access to child's device from everywhere by sending a command that locks the child's device remotely so it cannot be used until unlocked by a parent.
2. The application should offer the right to remote authorization or blocking access to certain elements by particular applications on the child's device by parents.
3. The application must notify the parent whenever some action with a remote control is activated, such as lock device or disable app.
4. The application must log any remote control operations carried out to refer to them at later times.

2.3.6 Module 6: Notifications and Alerts

Following are the requirements for Module 6:

1. Geofence alerts to be sent by the app in case the child leaves or returns the desired safe area.
2. The app will send screen time alerts to the parent if the child approaches the set daily or weekly screen time limit.
3. The app must alert the parent if the child breaches a predefined limit that outlines app restrictions and screen time.
4. The app must provide a notification history, allowing parents to review past alerts and notifications.

2.3.7 Module 7: Emergency SOS Feature

Following are the requirements for Module 7:

1. The app must provide an SOS button that, when pressed, sends an emergency signal with the child's GPS coordinates to the parent.
2. The location of the child must be automatically sent to the parent by a specified interval, say once every minute, upon activation of the SOS alarm.

3. Once the SOS signal is received by the application, it should quickly inform the parent of its receipt and thereafter display on the map the location of the child in real time.
4. The application should automatically cease sending the parent location updates after she reports the emergency or when the situation is solved.

2.4 Non-Functional Requirements

This section specifies nonfunctional requirements. These quality requirements should be specific, quantitative, and verifiable. The following are some examples of documenting guidelines.

2.4.1 Reliability

The system must ensure high reliability by minimizing the frequency and impact of failures. Reliability will be measured by the system's **Mean Time Between Failures (MTBF)** and its ability to detect, respond to, and recover from failures.

- **REL-1: Mean Time Between Failures (MTBF)**
  The system shall reach at least a minimum MTBF of **6** months under normal conditions of use. What this essentially means is that an app should rarely fail after more than every six months.
- **REL-2: Failure Classification and Response**
  The system must automatically detect and classify failures into two categories:
  - **Minor Failures**: Temporary issues that do not disrupt core functionalities, such as slight delays in notifications.
  - **Major Failures**: Critical issues where core functionalities (e.g., device locking, SOS alerts, real-time monitoring) are disrupted.

  For minor failures, the system must attempt to resolve the issue and retry the operation within **30 seconds**. For major failures, the system must immediately notify the parent and attempt to restart the affected services within **10 seconds**.
- **REL-3: Consequences of Failure**
  In the event of a system failure, the parent must be notified via a detailed error message explaining the nature of the failure and the steps being taken to resolve it. If the system is unable to recover from the failure within **1 hour**, the parent

should be provided with alternative means of control (e.g., remote manual control or notification via email/SMS).

2.4.2 Usability

The system must provide a user-friendly experience, ensuring that parents can easily navigate and use the app's features.

- **USE-1: Ease of Learning**
  The app should be easy to learn. First-time users (parents) should be able to accomplish simple tasks, such as setting screen time limits or blocking apps, within **10 minutes**. A tutorial or guide should be available to help new users understand the main features of the app.
- **USE-2: Ease of Use**
  All key features, including monitoring, tracking, and limiting usage, must be accessible directly from the home screen within **3 taps**. Buttons and labels must be clear, concise, and intuitive.
- **USE-3: Error Avoidance and Recovery**
  The app must confirm all critical operations, such as locking the child's device or deleting a geofence, to prevent errors. If an error occurs, the app must display a clear error message and offer a way to correct it. Additionally, the app should provide an option to **undo actions** (e.g., removing a restriction) within **5 seconds**.
- **USE-4: Efficiency of Interaction**
  The app should allow parents to perform important tasks efficiently. For example:
  - Parents should be able to set app restrictions within **1 minute**.
  - The app should display monitoring information within **5 seconds**.
  - The child's location should load in **less than 3 seconds**.

2.4.3 Performance

The system must meet specific performance criteria to ensure the smooth and timely operation of key features.

- **PER-1: Real-Time Monitoring Latency**
  The app must display the child's current screen activity within **3 seconds** of the parent initiating the real-time monitoring feature.

- **PER-2: Location Tracking Accuracy**
  The app must retrieve and display the child's location within **3 seconds** of the parent requesting it. The location data should be accurate within a **10-meter radius** under optimal GPS conditions.

- **PER-3: Geofencing Alert Speed**
  The system must notify the parent within **5 seconds** after the child crosses a geofenced boundary.

- **PER-4: App Restriction and Locking Response Time**
  The app must enforce app restrictions or lock the child's device within **5 seconds** of the parent initiating the restriction or lock command.

- **PER-5: Data Sync and Backup Frequency**
  Activity logs, keylogging data, and other important app data should be synchronized to the cloud server every **15 minutes**, and the backup process must complete within **2 minutes**.

- **PER-6: Notifications and Alerts Delivery**
  Notifications, such as exceeding screen time limits or SOS alerts, must be delivered to the parent's device within **5 seconds** of the event trigger.

- **PER-7: Application Loading Time**
  The app must load completely and be ready for use within **5 seconds** of being launched.

2.4.4 Security

The system must ensure the security of both the application and the data it processes, providing robust protection against unauthorized access.

- **SEC-1: User Authentication**
  The system should allow access to the app and its functions only to authorized users, such as parents. In the event of multiple failed login attempts, the system should lock the user's account after **5 failed login attempts**.

- **SEC-2: Data Protection**
  Restrictions must be in place for the storage and transmission of sensitive information, such as the child's location, activity logs, and keystroke data. Data must not be altered or accessed without proper authorization. If an unauthorized user attempts to access the data, a security alert must be triggered within **5-7 seconds**.

- **SEC-3: Data Access Controls**
  Data related to a child must only be accessible by the authenticated parent. Any attempt to access data outside the authorized user's scope must be logged and flagged for review. Any security breach must trigger an alert to the user and system administrators within **5 seconds** of detection.

- **SEC-4: Data Backup Security**
  Backup data, including activity logs and keystroke logs, must be encrypted and securely stored to prevent unauthorized access. Access to the backup should require a high level of authentication, and any attempt to tamper with or delete backup data must trigger an alert within **5 seconds**.

2.5 Domain Model

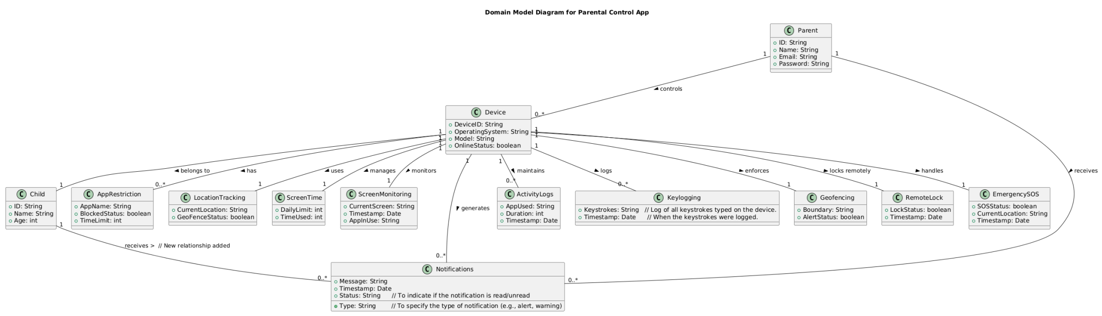

Figure 2.2: Domain Model

Chapter 3

System Overview

The Parental Control App is a feature-rich mobile application designed to empower parents to efficiently monitor and control their kids' gadget usage. By offering real-time monitoring, location tracking, app limits, screen time management, remote controls, notifications, alarms, and an emergency SOS feature, this app aims to create a safer online environment. The app provides a user-friendly interface for easy control and engagement, catering to the needs of both parents and children.

3.1 Functionality

The primary functionalities of the Parental Control App include:

- **Real-time Device Monitoring**: Parents can view real-time data regarding their child's device usage, including application usage statistics and battery status.
- **Location Tracking**: The app allows parents to track their child's real-time location using GPS data, providing peace of mind about their child's whereabouts.
- **App Restrictions**: Parents can set restrictions on specific applications, ensuring that their children are not accessing inappropriate or distracting content.
- **Screen Time Management**: The application enables parents to monitor and limit the time their children spend on devices, promoting healthy usage habits.
- **Remote Controls**: Parents can send remote commands to lock or unlock their child's device, providing a way to enforce screen time limits or intervene in emergencies.
- **Notifications and Alerts**: The app sends notifications to parents regarding important events, such as geofence breaches or unauthorized app usage.

- **Emergency SOS Feature**: In case of an emergency, children can trigger an SOS alert, which notifies their parents of their location and status, ensuring immediate assistance.

3.2 Context

In today's digital age, children are more exposed to various online dangers, such as inappropriate content, cyberbullying, and excessive screen time. To alleviate these concerns, the Parental Control App provides parents with the tools to monitor their kids' online activities and set appropriate limits. Thanks to the app's user-friendly design, parents can keep an eye on their children's device usage without invading their privacy.

3.3 Design

The design of the Parental Control App is centered around a modular architecture, allowing for easy updates and maintenance. The app is divided into several key modules:

- **Location Tracking Module**: Utilizes GPS technology to provide real-time location data.
- **Monitoring Module**: Gathers usage statistics and device status in real-time.
- **Restrictions Module**: Allows parents to enforce app restrictions and screen time limits.
- **Remote Control Module**: Facilitates remote commands sent to the child's device.
- **Notification System**: Handles alerts and notifications for parents.

These modules communicate with each other and with a backend service (Firebase/Google Maps API), ensuring seamless data flow and a cohesive user experience. The app prioritizes user privacy and data security, employing encryption and secure data storage practices to protect sensitive information.

3.4 Architectural Design

The architecture of the Parental Control App follows a **three-tier architecture** model, which divides the application into three distinct layers: the presentation layer, application layer, and data layer. This structure enhances scalability, maintainability, and separation of concerns within the system.

3.4.1 Presentation Layer

This layer is responsible for user interaction and is built as a mobile application, providing a user-friendly interface for both parents and children. It includes the following functionalities:

- **User Interface (UI)**: The mobile app offers parents an intuitive interface to monitor and control their child's device usage, and for children to view their app restrictions or trigger the SOS feature.
- **Device Control**: Parents can set restrictions, track the child's location, and manage screen time. The child receives notifications and can trigger the SOS feature in emergencies.
- **Communication with Business Layer**: The mobile app communicates with the backend server (application layer) via secure RESTful APIs to fetch and update data such as location, screen time limits, and app restrictions.

3.4.2 Application Layer

The application layer handles all the core processing and decision-making. It acts as an intermediary between the presentation layer and the data layer, managing business rules and enforcing security. Key components include:

- **Location Tracking Service**: Utilizes GPS and integrates with Google Maps APIs to provide real-time location tracking.
- **App Restriction Management**: Manages app restrictions and screen time limits set by parents and enforces them on the child's device.
- **SOS Notification System**: Handles SOS alerts triggered by the child and notifies the parents immediately, providing real-time location data.
- **Remote Control Commands**: Processes and forwards commands such as locking/unlocking the device and disabling specific apps.
- **Geofencing Logic**: Monitors geofence boundaries and sends alerts to parents when the child crosses the predefined area.

3.4.3 Data Layer

The data layer is responsible for securely storing all information related to the app, including user data, location history, app usage logs, and device restrictions. It interacts with external services and the database:

- **Database**: Stores user credentials, parental control settings, location history, and usage data. This information is stored securely and accessed by the application layer.
- **External APIs**: The data layer integrates with external services like Firebase for user authentication and Google Maps APIs for real-time location tracking.
- **Data Security**: Ensures that sensitive data such as child location and user credentials are encrypted and securely stored.

3.4.4 Architectural Pattern

The three-tier architecture pattern ensures a clear separation of responsibilities across different layers of the system:

- **Presentation Layer**: Focuses on user interaction and sending/receiving requests.
- **application layer**: Processes requests, applies business rules, and handles communication between the client-side and data-side.
- **Data Layer**: Stores and manages all persistent data, ensuring security and privacy.

This separation provides modularity, which allows for easier updates, testing, and debugging, as changes made in one layer do not necessarily affect the others.

3.4.5 Interaction Between Layers

Each layer communicates through well-defined interfaces:

- The **presentation layer** sends requests (such as setting restrictions or fetching location data) to the **application layer**.
- The **application layer** then applies the appropriate business rules (e.g., checking geofence, enforcing app restrictions) and retrieves/stores necessary data from the **data layer**.
- The **data layer** interacts with the database to store persistent data like location history, app usage logs, and restrictions.

3.4.6 Architectural Diagram

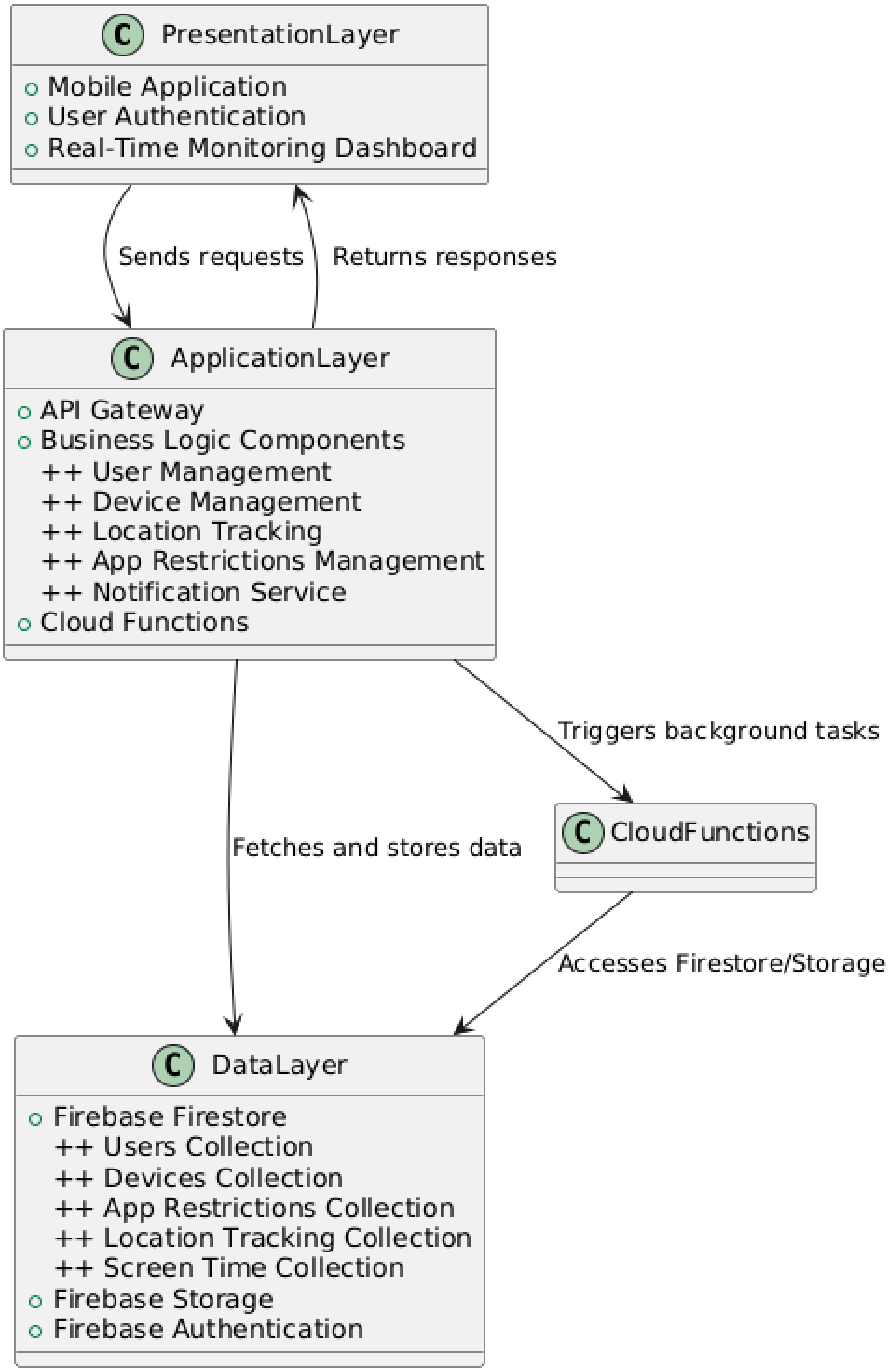

Figure 3.1: Three-Tier Architecture

This three-tier architectural design enhances the maintainability, scalability, and security of the Parental Control App, ensuring that the system remains responsive and robust, even as the user base grows.

3.5 Design Models

Design Models for Object Oriented Development Approach

3.5.1 Activity Diagram

The applicable models for the project using the object-oriented development approach include:

3.5.1.1 Activity Diagram

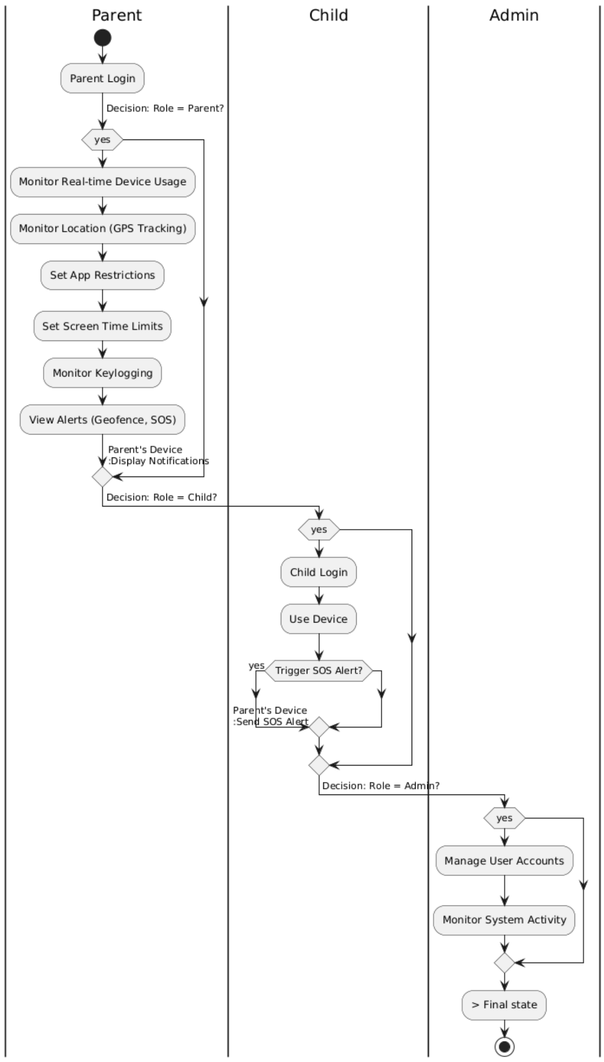

Figure 3.2: Activity Diagram

3.5.2 Class Diagram

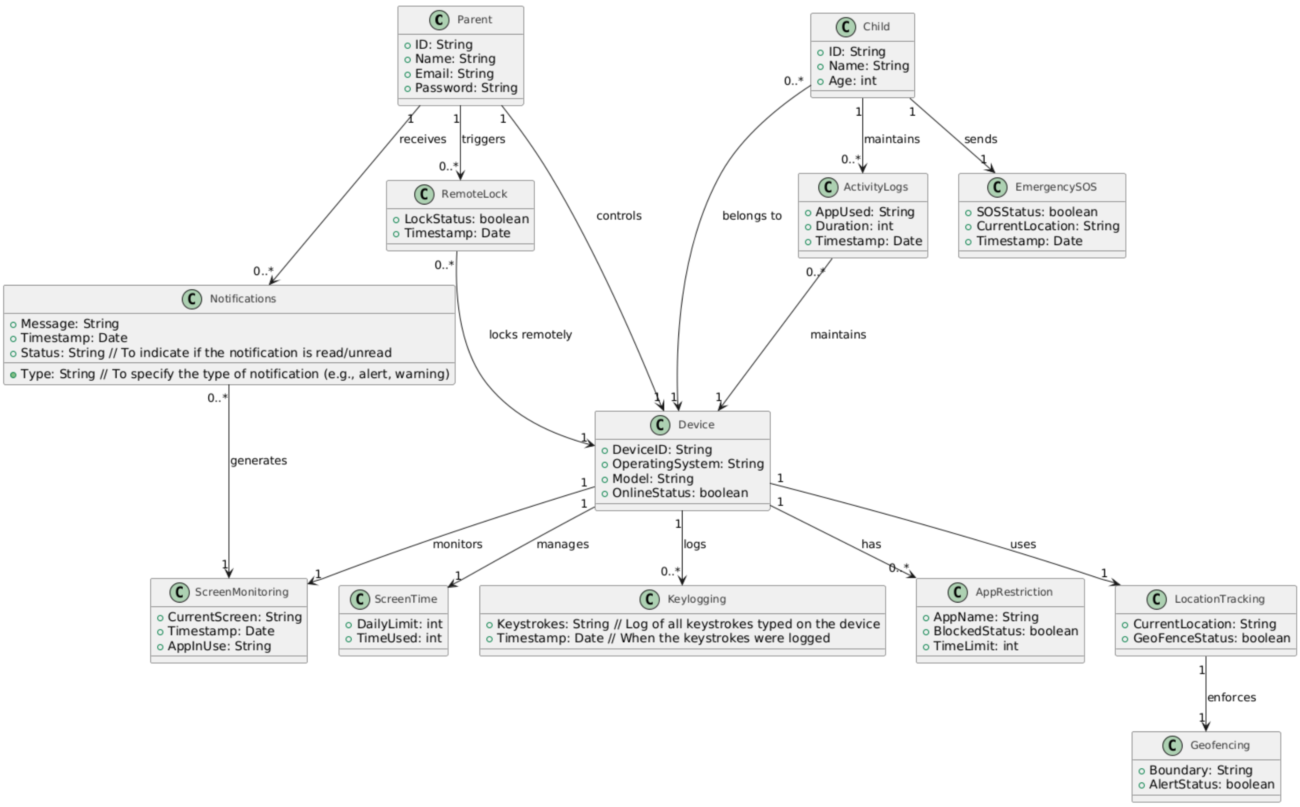

Figure 3.3: Class Diagram

3.5.3 Sequence Diagrams

3.5.3.1 Use Case 1: Monitor Child's Screen (Real-Time Monitoring)

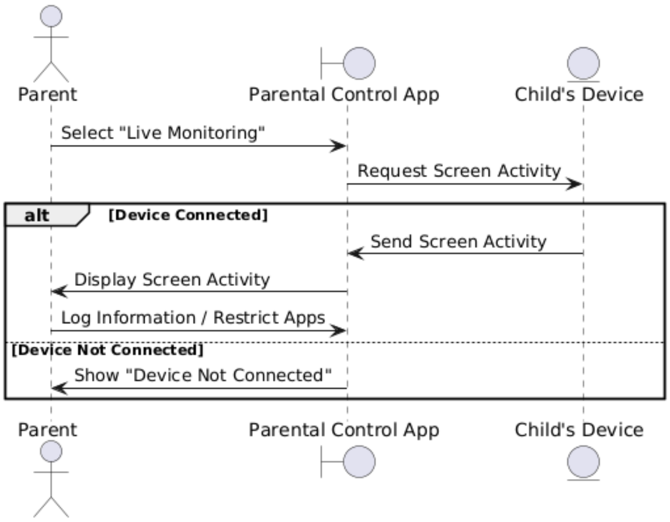

Figure 3.4: Sequence Diagram: Monitor Child's Screen

3.5.3.2 Use Case 2: Set App Restrictions

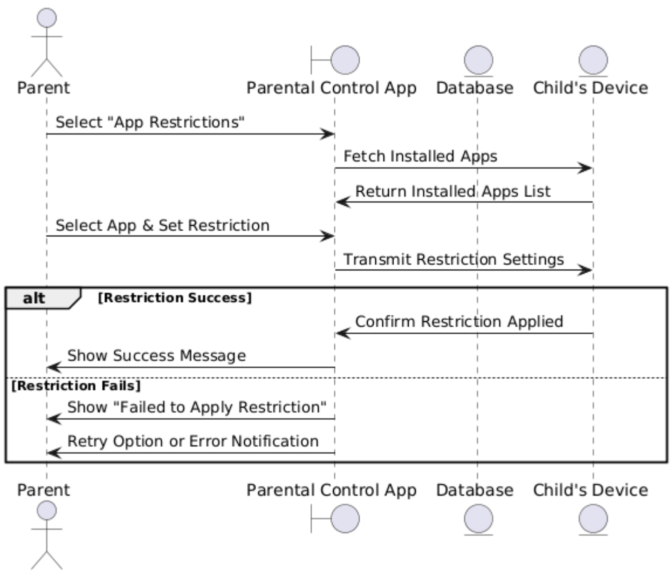

Figure 3.5: Sequence Diagram: Set App Restrictions

3.5.3.3 Use Case 3: Set Screen Time Limits

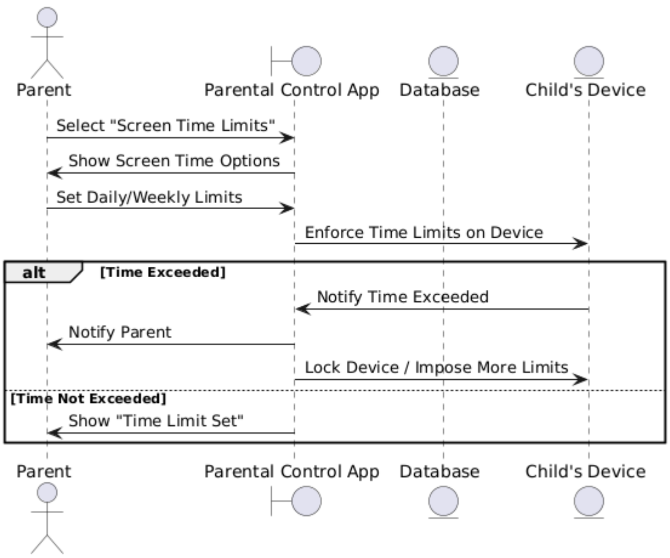

Figure 3.6: Sequence Diagram: Set Screen Time Limits

3.5.3.4 Use Case 4: Track Child's Location

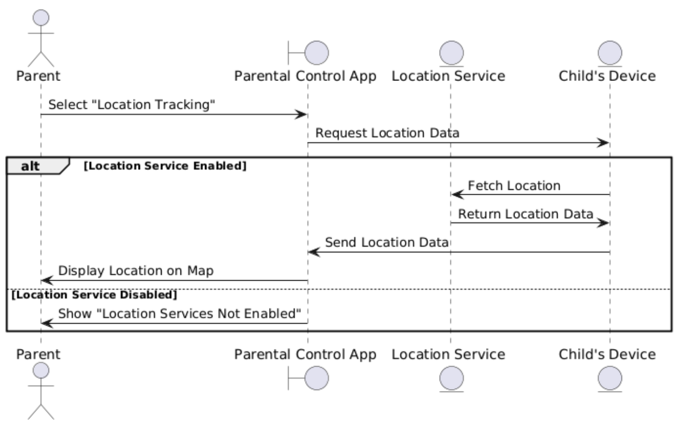

Figure 3.7: Sequence Diagram: Track Child's Location

3.5.3.5 Use Case 5: Set Geofencing Boundaries

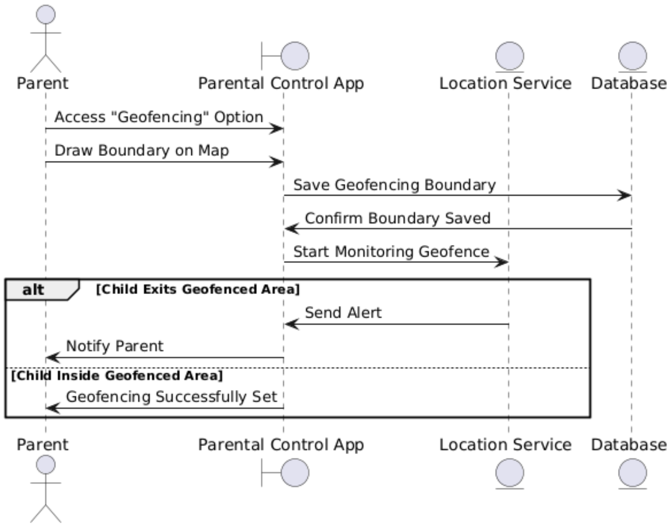

Figure 3.8: Sequence Diagram: Set Geofencing Boundaries

3.5.3.6 Use Case 6: Receive Alerts and Notifications

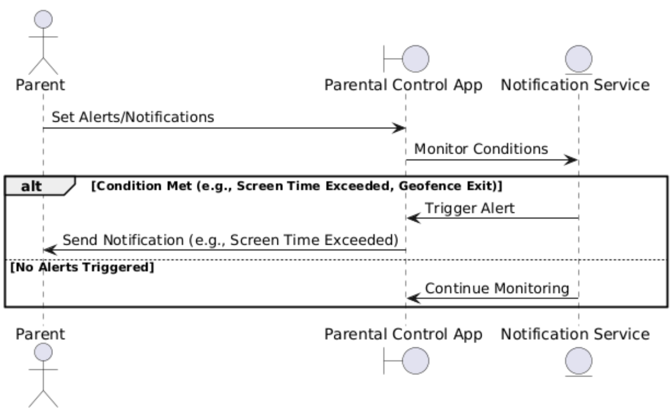

Figure 3.9: Sequence Diagram: Receive Alerts and Notifications

3.5.3.7 Use Case 7: Remote Lock Device

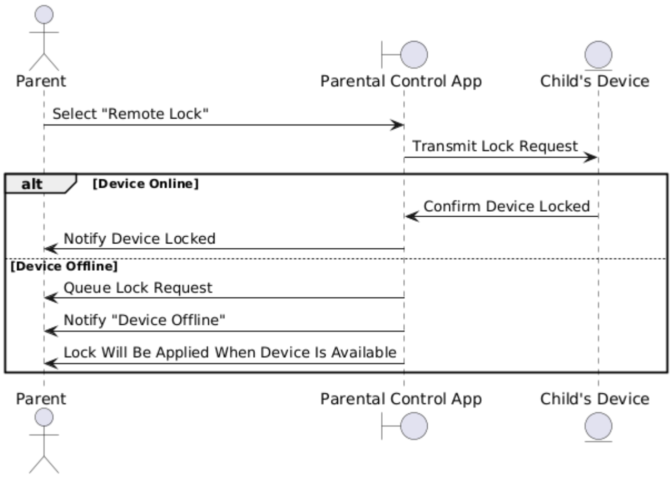

Figure 3.10: Sequence Diagram: Remote Lock Device

3.5.3.8 Use Case 8: Emergency SOS

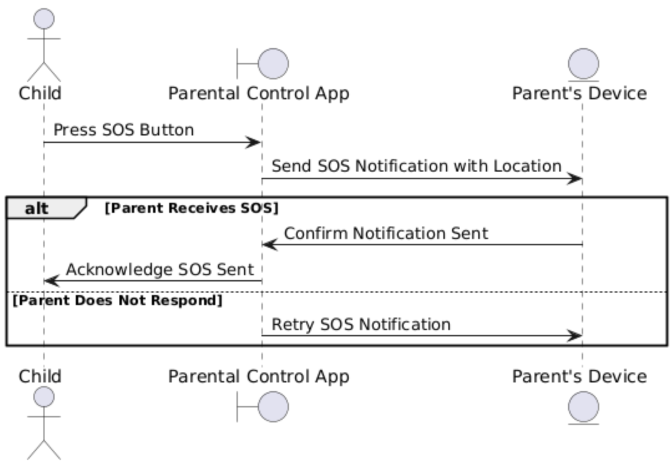

Figure 3.11: Sequence Diagram: Emergency SOS

3.5.3.9 Use Case 9: Monitor Keystrokes and Activity Logs

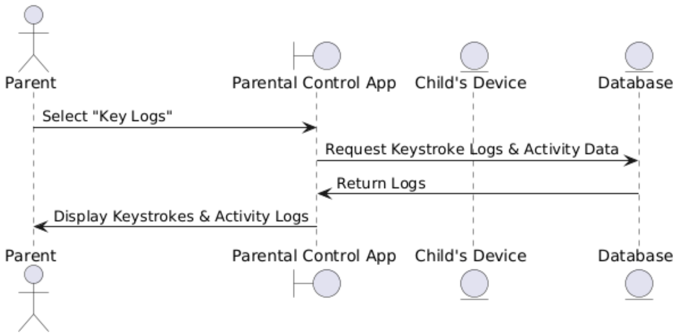

Figure 3.12: Sequence Diagram: Monitor Keystrokes and Activity Logs

3.5.4 State Transition Diagram

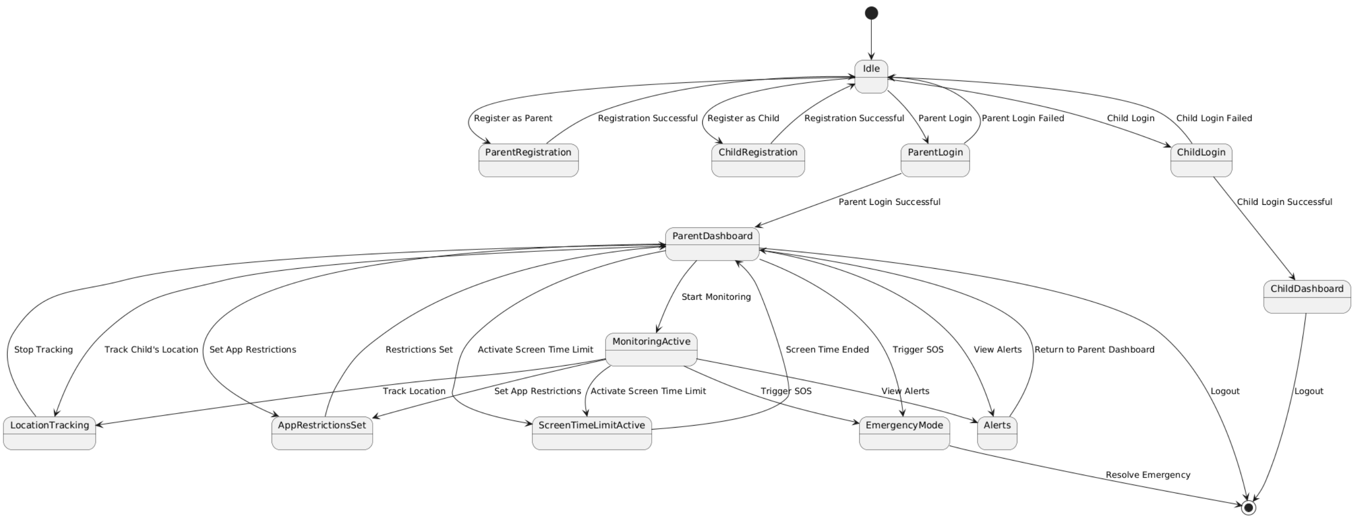

Figure 3.13: State Transition Diagram

3.6 Data Design

The information domain of the Parental Control App is transformed into well-structured data entities to ensure efficient management and monitoring. Each key entity within the system is mapped into data structures that handle the storage, processing, and organization of information.

3.6.1 Transformation of Information Domain into Data Structures

The system focuses on several core data entities, each representing a major aspect of the application:

- **User (Parent/Child):** Represents both parents and children using the system. Parents are responsible for managing restrictions, while children are tracked for device usage and location.
- **Device:** Represents the child's device that is monitored and controlled by the parent.
- **Location:** Tracks the real-time location of the child's device through GPS coordinates.
- **App Restrictions:** Manages restrictions set by the parent on specific apps.
- **Screen Time:** Monitors and limits the amount of time the child can spend on their device.
- **Notifications:** Sends alerts and updates to the parent about the child's activities.
- **Progress Report:** Generates reports based on the child's device usage and activities.

3.6.2 Data Storage and Processing

Each entity is stored in Firebase, a cloud-based NoSQL database provided by Google Cloud. Firebase uses collections and documents to represent the data structure. Each major system entity is represented as a collection with individual documents storing data related to the entity. Documents within collections are linked to one another via unique identifiers, ensuring efficient organization and processing.

- **User Data:** Stored in the `Users` collection, where each user document includes details such as name, email, role (parent or child), and other related metadata.
- **Device Data:** Stored in the `Devices` collection, where each device document includes the `userId` of the child who is using the device, and information such as device ID, device name, and lock status.
- **Location Data:** Stored in the `Locations` collection, tracking GPS coordinates in real-time, linked to the specific `deviceId`.

- **App Restrictions:** Stored in the `AppRestrictions` collection, where each document represents restrictions set for a particular app by the parent.
- **Screen Time Data:** Stored in the `ScreenTime` collection, where each document is linked to a specific `deviceId` and stores information on allowed screen time and usage statistics.
- **Notifications:** Stored in the `Notifications` collection, where notifications and alerts related to the child's activity are saved and sent to the parent.
- **Progress Reports:** Stored in the `ProgressReports` collection, where reports of a child's activities are generated and stored for the parent to review.

3.6.3 Data Organization and Databases

The system utilizes Firebase, a NoSQL, cloud-based database, for storing, processing, and organizing data. Firebase Firestore is used to structure the information domain into collections and documents, which are optimized for real-time synchronization and scalability.

3.6.4 Data Structure

Firebase uses a hierarchical data model, where:

- **Collections** act like tables (but schema-less).
- **Documents** within collections store data in key-value pairs (similar to rows in relational databases).
- Each document can contain sub-collections, making it flexible for nested data.

For example:

- **Users**: Parent and child accounts are stored as documents within the `users` collection, where parent and child data are distinguished by roles.
- **Devices**: Child devices are tracked as part of a `devices` collection, linked to the child's user ID.
- **App Restrictions**: Restrictions set by the parent are saved under the `app_restrictions` collection, each document containing details of the specific app and its usage rules.
- **Location Tracking**: The system stores location data in a `location_tracking` collection, with real-time updates reflecting the child's location.

3.6.5 Data Processing

Firebase Firestore allows for real-time updates and triggers using **Firebase Cloud Functions**. These functions automatically handle tasks such as:

- Enforcing screen time limits.
- Sending real-time notifications to parents when app restrictions are violated or a child enters/exits geofenced areas.
- Triggering SOS alerts and responding to emergency notifications.

Data is accessed via Firebase's SDKs, enabling real-time synchronization across multiple devices.

3.6.6 Data Storage

- **Firebase Firestore** handles structured data (app restrictions, screen time, location logs, notifications).
- **Firebase Storage** is used for media files such as screenshots, logs, or reports generated during device monitoring.
- **Firebase Authentication** securely manages user credentials (parents and children), ensuring role-based access to data.

3.6.7 Security and Privacy

Firebase provides powerful security features such as **Firebase Security Rules**, which define access permissions on a per-user basis. Data at rest and in transit is encrypted, adhering to industry-standard privacy guidelines. The system also applies role-based permissions to ensure parents can access children's activity logs, while children's data remains protected from unauthorized access.

3.6.8 Data Diagram

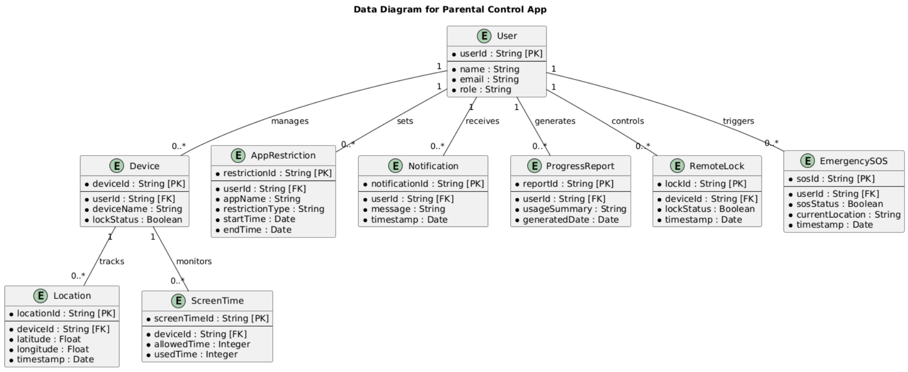

Figure 3.14: Data Diagram for the Parental Control App.

Description of the Data Diagram

The ERD for the Parental Control App represents the main entities within the system, structured to ensure efficient data management. Here's an overview of each entity and its relationships:

- **User**: Represents both parent and child users of the app. Parents can manage child accounts and receive notifications.
  - User has a one-to-many relationship with Device, as each parent can monitor multiple child devices.
  - User is associated with multiple AppRestriction, Notification, and Progress-Report entities, as they set restrictions, receive notifications, and review progress reports.
- **Device**: Represents the child's device, which is monitored by the parent.
  - Device is linked to Location for tracking real-time location.
  - Device also monitors ScreenTime, KeyLogging, and RemoteLock data, ensuring the child's activity is tracked, and devices can be locked remotely.
- **Location**: Tracks the real-time GPS location of the child's device.
  - Location is associated with Device and stores latitude, longitude, and timestamp data for location tracking.
- **AppRestriction**: Manages restrictions that the parent sets on specific apps on the child's device.

  - AppRestriction is linked to the User and stores information like app name, block status, and time limit.

- **ScreenTime**: Monitors the amount of time the child spends on their device, with daily limits and usage details.
  - ScreenTime is associated with Device and is used to track the child's screen time.

- **Notification**: Represents alerts or updates sent to the parent, providing information about the child's activity.
  - Notification is linked to the User and contains message, timestamp, status, and type details.

- **ProgressReport**: Generates and stores reports on the child's device usage and activities.
  - ProgressReport is associated with both User and Device, summarizing the child's device activity over a specified period.

- **KeyLogging**: Logs all keystrokes made on the child's device.
  - KeyLogging is linked to Device and keeps a log of keystrokes with timestamps for monitoring purposes.

- **RemoteLock**: Provides the ability for the parent to lock the child's device remotely.
  - RemoteLock is associated with Device and stores lock status and timestamp information.

- **EmergencySOS**: Handles SOS alerts and responds to emergency notifications.
  - EmergencySOS is associated with User and contains SOS status, current location, and timestamp information.

Bibliography

[1] A Kolyshkin and S Nazarovs. Stability of slowly diverging flows in shallow water. *Mathematical Modeling and Analysis*, 2007.